

Secure and Globally Distributed Static Website on AWS Using Terraform

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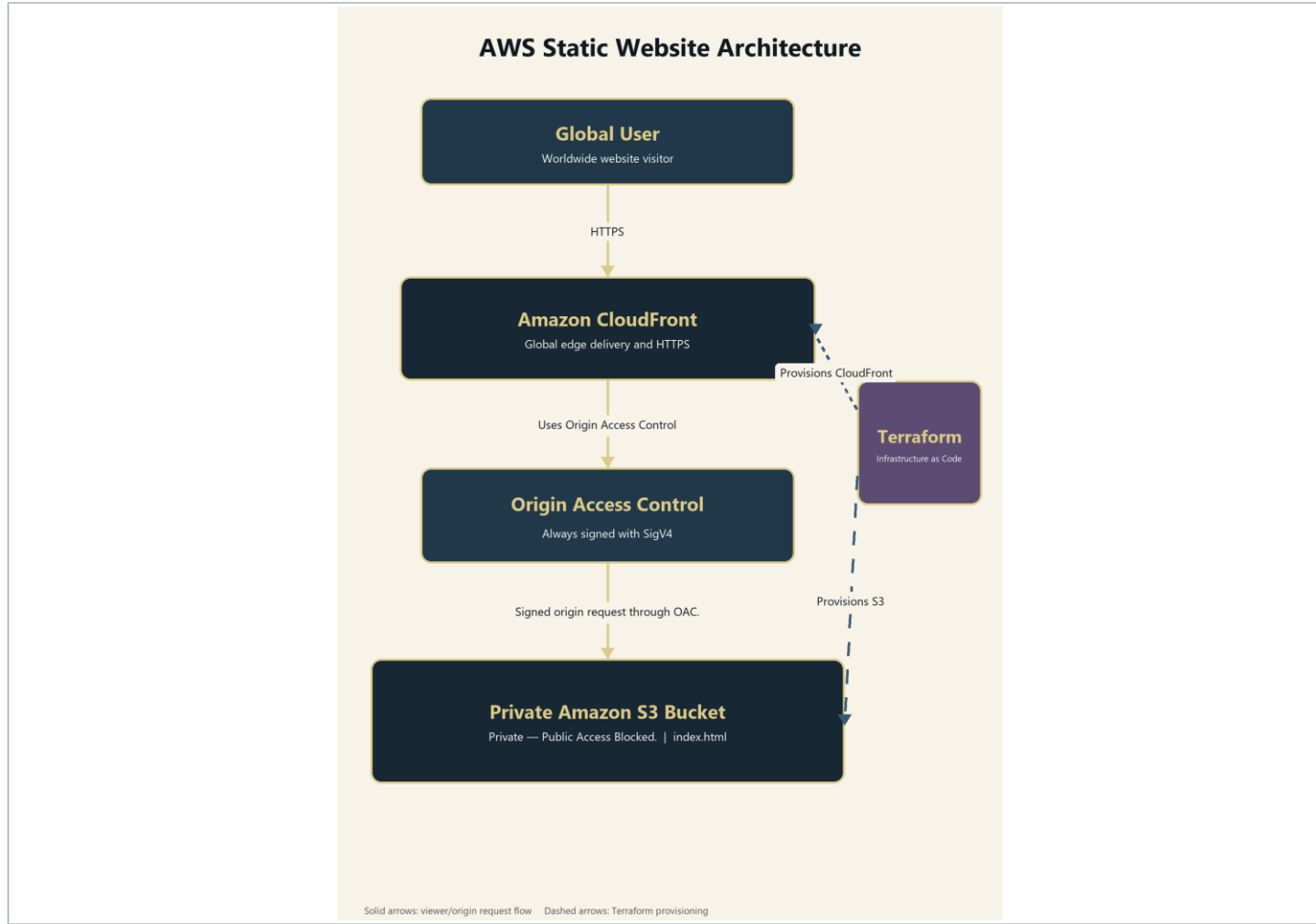
Static delivery still demands production quality

The challenge

Publish one self-contained HTML page worldwide without operating servers or exposing its storage origin.

01	HIGH AVAILABILITY	Managed, multi-AZ origin
02	GLOBAL DELIVERY	CloudFront edge network
03	AUTOMATIC SCALE	No fixed server capacity
04	SECURITY	HTTPS + private S3 origin
05	REPRODUCIBILITY	Terraform-defined lifecycle

CloudFront serves users while S3 stays private



REQUEST PATH

- 1 User requests the CloudFront URL over HTTPS.
- 2 CloudFront serves a cached copy or signs an origin request through OAC.
- 3 S3 returns index.html only to the designated distribution.

PROVISIONING FLOW

Terraform creates and links S3, OAC, CloudFront and the restricted bucket policy.

Figure 1. Request flow (solid) and Terraform provisioning (dashed).

Four managed services meet the requirement

S3

Durable managed storage for one static object.

CloudFront

Global HTTPS delivery, caching and compression.

OAC

SigV4 origin requests and distribution-scoped access.

Terraform

Versioned, reviewable and repeatable infrastructure.

REJECTED ALTERNATIVES

EC2 + load balancer add operating systems, patching, compute targets and cost without providing value to a static one-object site. ECS, Lambda, API Gateway and databases are also unnecessary.

Terraform makes the architecture repeatable

FILE STRUCTURE

```
main.tf
variables.tf
outputs.tf
.terraform.lock.hcl
site/
└─ index.html
diagrams/
└─ architecture-diagram.png
```

WORKFLOW

- | | | |
|-----------|---------------------------------|---------------------------|
| 01 | terraform init | Provider + lock file |
| 02 | terraform fmt / validate | Syntax and consistency |
| 03 | terraform plan | Review expected resources |
| 04 | terraform apply | Create or update AWS |

Stable first draft

<https://github.com/MahmoudBKanaan/AWS-Deployment/tree/portfolio-part-2-first-draft/iu-cloud-programming>

Security is enforced at every boundary

1	VIEWER	HTTP redirects to HTTPS	CloudFront default TLS certificate
2	EDGE → ORIGIN	OAC signs every S3 request	Bucket policy requires matching AWS:SourceArn
3	STORAGE	All four Public Access Block settings enabled	BucketOwnerEnforced · AES256 · versioning
4	SOURCE	No credentials in code or outputs	State, plans, keys and .env excluded from Git

Effective result: CloudFront is the only permitted reader of index.html.

Managed services remove capacity limits

01

S3 ORIGIN

Multi-AZ managed storage

No web server fleet

02

CLOUDFRONT EDGE

Worldwide points of presence

Low-latency delivery

03

CACHING

Managed-CachingOptimized

Fewer origin requests

04

SCALING

AWS-managed capacity

No capacity planning

The architecture scales by service design—not by forecasting server capacity.

Deployment evidence confirms the design

```
PowerShell - IU Cloud Programming

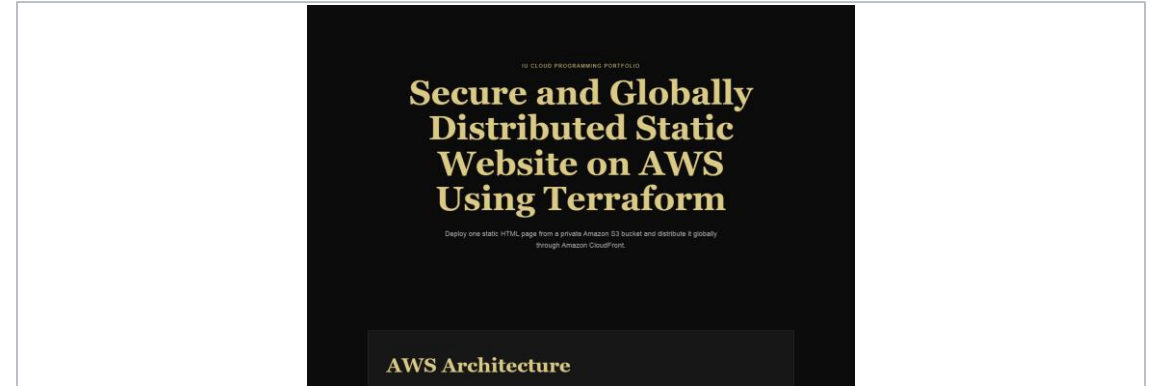
PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> terraform apply tfplan
aws_cloudfront_distribution.site: Creation complete after 2m34s [id=EZW53YPRFUFEA]
data.aws_iam_policy_document.allow_cloudfront: Read complete after 0s [id=377510756]
aws_s3_bucket_policy.allow_cloudfront: Creation complete after 0s
[id=iu-cloud-programming-spa-943865900195]

Apply complete! Resources: 9 added, 0 changed, 0 destroyed.

Outputs:

cloudfront_distribution_id = "EZW53YPRFUFEA"
cloudfront_domain_name = "d216eikm4r4k38.cloudfront.net"
s3_bucket_name = "iu-cloud-programming-spa-943865900195"
s3_origin_domain = "iu-cloud-programming-spa-943865900195.s3.eu-central-1.amazonaws.com"
website_url = "https://d216eikm4r4k38.cloudfront.net"
```

Terraform apply: 9 resources created.



Live website served through CloudFront HTTPS.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws cloudfront get-distribution --id
EZW53YPRFUFEA --profile iu-cloud --query
"Distribution.{Id:Id,Status:Status,DomainName:DomainName,Enabled:DistributionConfig.Enabled}" --output
json
{
  "Id": "EZW53YPRFUFEA",
  "Status": "Deployed",
  "DomainName": "d216eikm4r4k38.cloudfront.net",
  "Enabled": true
}
```

CloudFront distribution status: Deployed.

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> aws s3api head-object --bucket
iu-cloud-programming-spa-943865900195 --key index.html --profile iu-cloud --query "{Key:'index.html',C
ontentLength:ContentLength,ContentType:ContentType,CacheControl:CacheControl,ServerSideEncryption:Serv
erSideEncryption}" --output json
{
  "Key": "index.html",
  "ContentLength": 5874,
  "ContentType": "text/html; charset=utf-8",
  "CacheControl": "public, max-age=300",
  "ServerSideEncryption": "AES256"
}
```

S3 object metadata confirms HTML, cache and encryption.

Tests confirm security and repeatability

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> curl.exe -I $(terraform output -raw website_url)
HTTP/1.1 200 OK
Content-Type: text/html; charset=utf-8
Content-Length: 5874
Connection: keep-alive
Date: Fri, 17 Jul 2026 20:04:17 GMT
Last-Modified: Fri, 17 Jul 2026 19:38:17 GMT
ETag: "77767b45d4b4f13b0a8e8f39d471"
x-amz-server-side-encryption: AES256
Cache-Control: public, max-age=3600
x-amz-version-id: o5a_d1u7u8u0u1baf0tZ3v0u_Ada
Accept-Ranges: bytes
Server: AmazonS3
X-Cache: Hit from cloudfront
Via: 1.1 144487892b0bdf93d02a2d761e02f9.cloudfront.net (CloudFront)
X-Amz-CF-Ray: 3026-PS
X-Amz-CF-Id: cVqVY1i09dLUK9MTFqQUPVAb0Lm1l7g5G8AC113E0A-124RUw==
Age: 151
```

HTTPS 200 + CloudFront headers

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> curl.exe -I https://$(terraform output -raw s3_origin_domain)/index.html
HTTP/1.1 403 Forbidden
x-amz-request-id: 06QKwASBV2E9XNDW
x-amz-id-2: 8eGwda0SHMrDQMy1b4gVOEH0PF53Rky1CMeVU0j9hQf1ISARIPc+XchQCYE94o0K9qS8veYyAs9k7Va45u3VtqNo3N
jeM4lF(XInXbTdIY=
Content-Type: application/xml
Transfer-Encoding: chunked
Date: Fri, 17 Jul 2026 20:06:49 GMT
Server: AmazonS3
```

Direct S3 request denied (403)

```
PowerShell - IU Cloud Programming

PS D:\Documents\IU\Projects\CloudDep\iu-cloud-programming> 1..4 | ForEach-Object { curl.exe -sS -D - -o NUL $(terraform output -raw website_url) }
Request 1 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 285
Request 2 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
Request 3 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
Request 4 | HTTP/1.1 200 OK | X-Cache: Hit from cloudfront | Age: 286
```

Repeated request: Hit from CloudFront

TEST OUTCOMES

- HTTPS response: PASS
- HTTP → HTTPS redirect: PASS
- Direct S3 denial: PASS
- Public Access Block: PASS
- CloudFront cache hit: PASS
- Terraform no-change plan: PASS

Reflection

The strongest evidence combines positive delivery tests with negative origin-access tests and a final idempotency check.

The architecture meets every core requirement

RESULT

A globally accessible HTTPS site backed by a private S3 origin and reproducible Terraform configuration.

LIMITATIONS

Default CloudFront domain; no WAF, CI/CD or custom observability; designed for one static page.

FUTURE IMPROVEMENTS

Optional custom domain and ACM certificate, automated deployment, logging and budget alarms—only when justified by scope.

References

Amazon Web Services. Amazon CloudFront Developer Guide.
<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/Introduction.html>

Amazon Web Services. Restrict access to an Amazon S3 origin.
<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/private-content-restricting-access-to-s3.html>

Amazon Web Services. Data protection in Amazon S3.
<https://docs.aws.amazon.com/AmazonS3/latest/userguide/DataDurability.html>

HashiCorp. What is Terraform?
<https://developer.hashicorp.com/terraform/intro>

Code snapshot

<https://github.com/MahmoudBKanaan/AWS-Deployment/tree/portfolio-part-2-first-draft/iu-cloud-programming>